📘 Enterprise Resource Planning (ERP) — Complete Notes

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1. What is ERP?

Enterprise Resource Planning (ERP) is a type of integrated software system that organizations use to manage and automate core business processes across departments — including finance, HR, manufacturing, supply chain, procurement, sales, and more — through a single unified platform.

> ***In simple terms:*** *ERP is the "central nervous system" of a business. It connects all departments so they can share data, collaborate, and work efficiently in real time.*

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2. Why Was ERP Created? (Origin Story)

2.1 Pre-ERP Era — The Problem

Before ERP systems existed, businesses used **siloed software**:

- Finance had its own system
- Inventory had its own system
- HR had its own system
- These systems **did not talk to each other**

This caused:

- Duplicate data entry
- Data inconsistencies
- Delayed reporting
- Poor decision-making
- High operational costs

2.2 The Evolution Timeline

Era	System	Description
1960s	**Inventory Control**	Simple inventory management software
1970s	**MRP (Material Requirements Planning)**	Managed manufacturing materials and schedules
1980s	**MRP II (Manufacturing Resource Planning)**	Extended to include shop-floor and distribution
1990s	**ERP**	Gartner coined the term; full enterprise integration
2000s	**ERP II**	Web-based, extended to customers & partners
2010s	**Cloud ERP**	SaaS delivery model, mobile access
2020s	**Intelligent ERP**	AI, ML, IoT, real-time analytics integration

2.3 Who Coined "ERP"?

The term **ERP** was coined by the research firm **Gartner Group in 1990** to describe a new generation of systems that extended MRP II beyond manufacturing into enterprise-wide resource management.

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3. Why Do We Use ERP?

3.1 Core Reasons

1. **Integration** — All departments share one database; no data silos
2. **Real-time Information** — Managers get live dashboards and reports
3. **Process Automation** — Reduces manual work (e.g., invoice generation, payroll)
4. **Standardization** — Enforces consistent business processes across locations
5. **Compliance** — Helps meet regulatory and audit requirements (GST, SOX, GDPR)
6. **Cost Reduction** — Cuts IT costs by replacing multiple systems with one
7. **Scalability** — Grows with the business
8. **Better Decision-Making** — Centralized data improves analytics and forecasting

3.2 Business Problems ERP Solves

Problem	ERP Solution
Duplicate data	Single source of truth
Delayed reports	Real-time dashboards
Manual calculations	Automated workflows
Communication gaps	Integrated modules
Compliance risk	Built-in audit trails
Inventory errors	Live stock tracking

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4. How Does ERP Work?

4.1 Architecture — The Central Database Concept

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┌──────────┐   ┌──────────┐   ┌──────────┐   ┌──────────┐
| Finance  |   |  Sales   |   |   HR     |   |  Supply  |
| Module   |   | Module   |   | Module   |   |  Chain   |
└────┬─────┘   └────┬─────┘   └────┬─────┘   └────┬─────┘
     |              |              |              |
     └──────────────┴──────────────┴──────────────┘
                          |
                ┌─────────▼─────────┐
                |   CENTRAL DATABASE  |
                |   (Single Source    |
                |    of Truth)        |
                └─────────────────────┘
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All modules read and write to the **same central database**, ensuring data consistency across the entire organization.

4.2 Workflow Example

1. Sales team creates a **Sales Order**
2. ERP automatically checks **Inventory**
3. If stock is low, it triggers a **Purchase Order**
4. Goods are received and **Inventory updates**
5. **Invoice is generated** and sent to customer
6. Payment received updates **Financial Accounts**
7. All steps are tracked in **real-time reports**

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5. Core Modules of ERP

5.1 Financial Management (Finance Module)

- General Ledger
- Accounts Payable & Receivable
- Budgeting and Forecasting
- Fixed Assets Management
- Tax Management (GST, VAT, etc.)

5.2 Human Resource Management (HRM)

- Employee Records
- Payroll Processing
- Attendance & Leave Management

- Recruitment & Onboarding
- Performance Management
- Training and Development

5.3 Supply Chain Management (SCM)

- Procurement & Purchasing
- Supplier Management
- Logistics & Shipping
- Warehouse Management
- Demand Planning

5.4 Manufacturing / Production

- Bill of Materials (BOM)
- Production Planning & Scheduling
- Shop Floor Control
- Quality Management
- Machine Maintenance (MRO)

5.5 Sales & Customer Relationship Management (CRM)

- Lead & Opportunity Management
- Order Management
- Pricing & Discounts
- Customer Service
- Sales Forecasting

5.6 Inventory Management

- Stock Tracking (Real-time)
- Reorder Level Alerts
- Batch and Serial Number Tracking
- Multi-warehouse Support
- Stock Valuation (FIFO, LIFO, Weighted Average)

5.7 Project Management

- Project Planning & Tracking
- Resource Allocation
- Budgeting per Project
- Milestone Management

5.8 Business Intelligence (BI) & Reporting

- KPI Dashboards
- Custom Reports
- Predictive Analytics
- Data Visualization

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6. Types of ERP Systems

6.1 By Deployment Model

Type	Description	Examples
On-Premise ERP	Installed on company's own servers	SAP ECC, Oracle E-Business Suite
Cloud ERP (SaaS)	Hosted on vendor's cloud	SAP S/4HANA Cloud, NetSuite
Hybrid ERP	Mix of on-premise and cloud	SAP Rise with S/4HANA

6.2 By Business Size

Size	Type	Examples
Small Business	Lightweight ERP	Zoho ERP, Odoo
Mid-Market	Mid-range ERP	Microsoft Dynamics 365, Sage
Enterprise	Full-suite ERP	SAP S/4HANA, Oracle ERP Cloud

6.3 By Industry

- **Manufacturing ERP** — Infor LN, Epicor
- **Retail ERP** — Oracle Retail, SAP Retail
- **Healthcare ERP** — Infor Healthcare, Oracle Health
- **Construction ERP** — Procore, Viewpoint
- **Education ERP** — Ellucian, Unit4

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7. Major ERP Vendors & Products

Vendor	Product	Market Focus
SAP	SAP S/4HANA	Large Enterprises
Oracle	Oracle ERP Cloud / NetSuite	Large & Mid
Microsoft	Dynamics 365	Mid & Large
Infor	Infor CloudSuite	Industry-specific
Epicor	Epicor Kinetic	Manufacturing
Sage	Sage 300, Sage X3	SMEs

Vendor	Product	Market Focus
Odoo	Odoo ERP	SMEs (Open Source)
Zoho	Zoho ERP	Small Business
IFS	IFS Cloud	Asset-heavy Industries
Tally	TallyPrime	India SMEs

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8. ERP Versions & Evolution

8.1 Generation 1 — MRP (1960s–70s)

- Focused only on **inventory and production scheduling**
- IBM developed first MRP systems
- Tools: BOMP (Bill of Material Processor)

8.2 Generation 2 — MRP II (1980s)

- Extended MRP to include:
  - Capacity planning
  - Shop floor control
  - Financial planning
- Introduced by **Oliver Wight**
- Key term coined: "Manufacturing Resource Planning"

8.3 Generation 3 — ERP (1990s)

- Gartner coined "ERP" in 1990
- SAP R/3 launched in 1992 — landmark product
- Extended beyond manufacturing to all enterprise processes
- Client-server architecture
- Companies: SAP, Oracle, PeopleSoft, JD Edwards

8.4 Generation 4 — ERP II (2000s)

- Gartner introduced "ERP II" in 2000
- Extended to external stakeholders (customers, suppliers)
- Web-enabled interfaces
- CRM and SCM integrated
- E-commerce integration

8.5 Generation 5 — Cloud ERP (2010s)

- Shift from on-premise to SaaS delivery
- Subscription-based pricing
- Mobile accessibility
- Examples: NetSuite (first cloud ERP, launched 1998), SAP Business ByDesign

8.6 Generation 6 — Intelligent/Digital ERP (2020s)

- Embedded **Artificial Intelligence (AI)**
- **Machine Learning** for predictions and automation
- **IoT integration** for real-time manufacturing data
- **Robotic Process Automation (RPA)**
- Natural Language Processing for queries
- Examples: SAP S/4HANA with AI, Oracle Fusion with GenAI

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9. ERP Implementation

9.1 Implementation Approaches

Approach	Description	Risk
Big Bang	All modules go live at once	High
Phased Rollout	Module by module, department by department	Medium
Parallel Adoption	Old & new systems run side-by-side	Low but costly
Hybrid	Mix of above approaches	Balanced

9.2 Implementation Phases

1. **Project Planning** — Scope, timeline, team, budget
2. **Business Process Review (BPR)** — Map current vs. future processes
3. **System Design** — Configuration and customization
4. **Data Migration** — Cleanse and import legacy data
5. **Testing** — Unit, Integration, UAT (User Acceptance Testing)
6. **Training** — End-user and admin training
7. **Go-Live** — System launch
8. **Post Go-Live Support** — Bug fixes and optimization

9.3 Key Roles in ERP Implementation

- **Project Manager** — Coordinates the implementation
- **Functional Consultant** — Configures business processes

- **Technical Consultant** — Handles customizations and integrations
- **Change Manager** — Manages people/organizational change
- **Data Migration Specialist** — Moves data from old systems
- **End Users / Key Users** — Business users who test and use the system

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10. Advantages & Disadvantages of ERP

10.1 Advantages ✅

- Single, unified database across the organization
- Eliminates data redundancy and errors
- Real-time visibility into all operations
- Improved collaboration across departments
- Better compliance and audit trails
- Reduced operational and IT costs (long-term)
- Scalable and adaptable to business growth
- Supports better strategic planning

10.2 Disadvantages ❌

- **High Implementation Cost** — Licensing, consulting, hardware
- **Long Implementation Time** — Months to years
- **Complexity** — Requires heavy training
- **Customization Risk** — Over-customization makes upgrades difficult
- **Change Resistance** — Employees resist new workflows
- **Vendor Lock-in** — Switching ERPs is expensive
- **Data Migration Challenges** — Legacy data is often dirty
- **Business Disruption** — Transitional period can affect operations

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11. ERP vs Other Systems

Feature	ERP	CRM	SCM
Scope	Entire enterprise	Customer relations	Supply chain
Focus	All operations	Sales & marketing	Procurement & logistics
Data	All business data	Customer data	Vendor/inventory data
Integration	Complete	Partial	Partial
Example	SAP, Oracle	Salesforce	SAP Ariba

> *Note: Modern ERP systems often **include CRM and SCM modules**, so the lines are blurring.*

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12. ERP in the Indian Context

12.1 ERP Adoption in India

- Growing rapidly due to **GST implementation (2017)** — forced businesses to digitize
- **Make in India** initiative boosted manufacturing ERP demand
- Popular in sectors: Manufacturing, FMCG, Pharma, IT Services, Retail

12.2 Popular ERP in India

- **SAP S/4HANA** — Large enterprises (Tata, Reliance, Infosys)
- **Oracle ERP Cloud** — BFSI and IT companies
- **Microsoft Dynamics 365** — Mid-market
- **Tally ERP / TallyPrime** — SMEs (most widely used for accounting)
- **Zoho ERP / Odoo** — Startups and SMEs

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13. Future of ERP

13.1 Emerging Trends

1. **AI & Generative AI** — Automated insights, chatbots for ERP queries
2. **Cloud-first Strategy** — 80%+ deployments moving to cloud by 2026
3. **Composable ERP** — Modular architecture (plug-and-play modules)
4. **Low-code / No-code Customization** — Business users can customize without developers
5. **Sustainability Modules** — Carbon tracking and ESG reporting
6. **Real-time Analytics** — In-memory computing (e.g., SAP HANA)
7. **IoT Integration** — Smart factory and Industry 4.0 compatibility
8. **Hyperautomation** — Combining AI, RPA, and ML for end-to-end automation

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14. Key ERP Terminology

Term	Definition
BOM	Bill of Materials — list of raw materials for production
MRP	Material Requirements Planning
WBS	Work Breakdown Structure — for project management
GL	General Ledger — master accounting record
AP/AR	Accounts Payable / Accounts Receivable
SKU	Stock Keeping Unit — unique inventory identifier
UAT	User Acceptance Testing — final testing by end users
RFP	Request for Proposal — document sent to ERP vendors

Term	Definition
SaaS	Software as a Service — cloud delivery model
FIFO/LIFO	First In First Out / Last In First Out — inventory valuation
KPI	Key Performance Indicator — measurable business metric
SOX	Sarbanes-Oxley Act — US financial compliance regulation
GST	Goods and Services Tax — Indian tax system
RPA	Robotic Process Automation — bot-based task automation

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15. Quick Revision Summary

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ERP = Enterprise Resource Planning

Origin:   Inventory Control (60s) → MRP (70s) → MRP II (80s) → ERP (90s) →
          ERP II (2000s) → Cloud ERP (2010s) → Intelligent ERP (2020s)

Purpose:  Integrate ALL business processes into ONE system

Modules:  Finance | HR | SCM | Manufacturing | Sales | Inventory | BI

Types:    On-Premise | Cloud (SaaS) | Hybrid

Top ERP:  SAP | Oracle | Microsoft Dynamics | Infor | Odoo | Tally

Benefits: Integration | Real-time data | Automation | Compliance | Cost savings

Challenges: Cost | Time | Complexity | Change management
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Notes compiled for complete ERP understanding — covers origin, evolution, usage, modules, vendors, implementation, and future trends.